



Curriculum

[Bulk Uploader](#)

Start putting together your course by creating sections, lectures and practice activities ([quizzes, coding exercises and assignments](#)). Use your [course outline](#) to structure your content and label your sections and lectures clearly. If you're intending to offer your course for free, the total length of video content must be less than 2 hours.

Section 1: Introduction

✓ Lecture 1: Introduction

(Preview enabled) ▾

Unpublished lecture: Course Update Notice

[+ Content](#) ^

[+ Description](#)

[+ Lab](#)

Unpublished lecture: What You Need Before You Start

[+ Content](#) ^

[+ Description](#)

[+ Lab](#)

[+ Curriculum item](#)

Unpublished Section: 📄 Introduction to Semantic Kernel

⚠️ Unpublished lecture: 📄 Section Overview

[+ Content](#)



Understand the topics covered in this foundational section.

⚠️ Unpublished lecture: 📄 What is an LLM?

[+ Content](#)



Understand what large language models are and how they are used in applications.

⚠️ Unpublished lecture: 📄 What is Generative AI?

[+ Content](#)



Distinguish generative AI from broader AI concepts and understand common usage patterns.

⚠️ Unpublished lecture: 📄 What is Semantic Kernel?

[+ Content](#)



⚠️ Unpublished lecture: 📄 Why Use Semantic Kernel Instead of Calling the...



[+ Content](#)

⚠️ Unpublished lecture: 📄 Semantic Kernel vs Microsoft Agent Framework



[+ Content](#)

⚠️ Unpublished lecture: 📄 Understanding AI Agents in Business Context



[+ Content](#)

Understand what AI agents are and where they fit in modern business solutions.

⚠️ Unpublished lecture: 📄 Section Review

[+ Content](#)



Reinforce the relationship between LLMs, generative AI, Semantic Kernel, and agents.

+ Curriculum item

Unpublished Section: 📄 Environment Setup and Starter Solution

⚠️ Unpublished lecture: 📄 Section Overview

+ Content



Understand the development tools, dependencies, and project structure.

⚠️ Unpublished lecture: 📄 Install .NET 10, Tools, and Extensions [Demo]



+ Content

Set up the required SDKs, IDE support, and local tooling.

⚠️ Unpublished lecture: 📄 Azure OpenAI Prerequisites and Resource Plan...



+ Content

Understand what Azure resources are needed and how model deployments fit into the solution.

⚠️ Unpublished lecture: 📄 Create the Starter Solution [Demo]



+ Content

Build the initial .NET 10 solution structure that will evolve through the course.

⚠️ Unpublished lecture: 📄 Configure Secrets and App Settings [Demo]



+ Content

Store endpoints, keys, and environment settings securely.

⚠️ Unpublished lecture: 📄 Verify the Baseline Application Runs [Demo]



+ Content

Confirm the learner has a working starting point before Semantic Kernel is added.

⚠ Unpublished lecture: 📄 Section Review

+ Content



Confirm the baseline environment is ready for development.

+ Curriculum item

Unpublished Section: 📄 Building Your First Kernel

⚠ Unpublished lecture: 📄 Section Overview

+ Content



Understand what the kernel is and how it fits inside the app.

⚠ Unpublished lecture: 📄 Understanding the Kernel, Services, and Conne...



+ Content

Explain the kernel as the central orchestrator in Semantic Kernel.

⚠ Unpublished lecture: 📄 Add Semantic Kernel to the Solution [Demo]



+ Content

Install and reference the core packages correctly.

⚠ Unpublished lecture: 📄 Register Azure OpenAI Chat Services [Demo]



+ Content

Connect the project to Azure OpenAI through Semantic Kernel.

⚠ Unpublished lecture: 📄 Build a First Chat Completion Console Flow [De...



[+ Content](#)

Send a prompt, receive a response, and understand the execution flow.

⚠ Unpublished lecture: 📄 Add Basic Chat History [Demo]

[+ Content](#)



Maintain short conversational context correctly.

⚠ Unpublished lecture: 📄 Refactor the Kernel Setup into Reusable Servic...



[+ Content](#)

Move from one-off code to maintainable application architecture.

⚠ Unpublished lecture: 📄 Section Review

[+ Content](#)



Confirm the baseline environment is ready for development.

[+ Curriculum item](#)

Unpublished Section: 📄 Semantic Kernel Plugins

⚠ Unpublished lecture: 📄 Section Overview

[+ Content](#)



Understand how plugins and prompts extend the capabilities of a kernel.

⚠ Unpublished lecture: 📄 Introduction to Plugins

[+ Content](#)



Understand what a plugin is and how the kernel invokes plugin functionality.

⚠ Unpublished lecture: 📄 Using Built-In Plugins [Demo]

[+ Content](#)



Learn how to use built-in plugin features, such as time and summarization.

⚠ Unpublished lecture: 📄 Prompt Engineering for Application Developers ^

+ Content

Learn how to write application-grade prompts instead of ad hoc chat prompts.

⚠ Unpublished lecture: 📄 Prompt Templates, Variables, and Arguments ^

+ Content

Understand how Semantic Kernel templating supports reusable instructions.

⚠ Unpublished lecture: 📄 Create Reusable Prompt Templates [Demo] ^

+ Content

Build prompt templates for project-specific assistant behavior.

⚠ Unpublished lecture: 📄 Add Personas and Role-Based Prompting [Demo] ^

+ Content

Shape assistant tone, domain behavior, and response constraints.

⚠ Unpublished lecture: 📄 Build a Semantic Prompt Plugin [Demo] ^

+ Content

Organize prompt behavior into a reusable semantic plugin.

⚠ Unpublished lecture: 📄 Store Prompts in Files and Organize Them Clea... ^

+ Content

Externalize prompts for maintainability and team collaboration.

⚠ Unpublished lecture: 📄 Test and Tune Prompt Quality [Demo] ^

+ Content

Compare outputs and refine prompts using practical criteria.

⚠ Unpublished lecture: 📄 Section Review

+ Content



Confirm the learner can create and organize prompt-based plugins effectively.

+ Curriculum item

Unpublished Section: 📄 Plugins and Native Functions

⚠ Unpublished lecture: 📄 Section Overview

+ Content



Understand how plugins extend AI behavior.

⚠ Unpublished lecture: 📄 Understanding Plugins, Functions, and Discover...



+ Content

Learn what makes a function usable by the model and by the kernel.

⚠ Unpublished lecture: 📄 Build the First Native Plugin [Demo]



+ Content

Create a plugin that exposes business logic to the kernel.

⚠ Unpublished lecture: 📄 Add Descriptions and Parameters That Improve ...



+ Content

Make plugin methods easier for the model to choose and call correctly.

⚠ Unpublished lecture: 📄 Use Built-In Plugins Where Appropriate [Demo]



+ Content

Apply built-in utilities without bloating custom code.

⚠ Unpublished lecture: 📄 Combine Prompt Templates with Plugin Calls [D...



+ Content

Blend instructions and executable functions in one flow.

⚠ Unpublished lecture: 📄 Use Functions Inside Prompts [Demo]



+ Content

Inject function output into prompt execution for smarter results.

⚠ Unpublished lecture: 📄 Automatically Invoking Functions

+ Content



Understand when and why automatic function calling is useful.

⚠ Unpublished lecture: 📄 Configure Automatic Function Invocation [Demo]



+ Content

Enable the kernel to choose and invoke registered functions automatically.

⚠ Unpublished lecture: 📄 Design Rules for Good Plugins

+ Content



Understand when logic should live in prompts, functions, or application code.

⚠ Unpublished lecture: 📄 Section Review

+ Content



Confirm the learner can build and register useful plugins.

+ Curriculum item

Unpublished Section: 📄 Moving into ASP.NET Core

⚠ Unpublished lecture: 📄 Section Overview

[+ Content](#)



Understand how the console logic moves into a web app.

⚠ Unpublished lecture: 📄 Create the ASP.NET Core MVC Starter Portal [D...



[+ Content](#)

Build the baseline business website that will host the embedded copilot.

⚠ Unpublished lecture: 📄 Add Semantic Kernel to ASP.NET Core Depend...



[+ Content](#)

Move kernel configuration into the ASP.NET Core composition root.

⚠ Unpublished lecture: 📄 Build a Semantic Kernel Service Layer [Demo]



[+ Content](#)

Separate kernel orchestration from controllers and UI code.

⚠ Unpublished lecture: 📄 Add the Documents Area to the Portal [Demo]



[+ Content](#)

Introduce a business-facing area where users can interact with managed content.

⚠ Unpublished lecture: 📄 Add the Requests Area to the Portal [Demo]



[+ Content](#)

Add another business context where AI assistance can provide value.

⚠ Unpublished lecture: 📄 Build an Embedded Copilot Panel [Demo]



[+ Content](#)

Add a contextual assistant panel to the website instead of building a standalone chat app.

⚠ Unpublished lecture: 📄 Pass Page Context into the Copilot [Demo]



+ Content

Make the assistant aware of the current document, request, or page context.

⚠ Unpublished lecture: 📄 Test the Embedded Copilot Experience [Demo]



+ Content

Verify that the assistant works as part of the portal workflow.

⚠ Unpublished lecture: 📄 Section Review

+ Content



Confirm the assistant now works inside ASP.NET Core.

+ Curriculum item

Unpublished Section: 📄 Contextual Assistance, History, and Grounding

⚠ Unpublished lecture: 📄 Section Overview

+ Content



Understand what makes an embedded copilot context-aware and useful.

⚠ Unpublished lecture: 📄 Persist Copilot Conversations [Demo]



+ Content

Store user and assistant interactions per portal context.

⚠ Unpublished lecture: 📄 Reload Historical Context into Semantic Kernel ...



+ Content

Rehydrate prior conversations for follow-up questions.

⚠ Unpublished lecture: 📄 Add Document Upload Support [Demo]



[+ Content](#)

Allow users to submit PDFs into the application workflow.

⚠ Unpublished lecture: 📄 Extract Text from PDFs with IronPDF [Demo]



[+ Content](#)

Pull machine-readable text from uploaded documents.

⚠ Unpublished lecture: 📄 Handle Scanned PDFs with OCR [Demo]



[+ Content](#)

Process image-based PDFs that do not already contain readable text.

⚠ Unpublished lecture: 📄 Build a Document Preprocessing Pipeline [Demo]



[+ Content](#)

Move from raw PDF text extraction into structured ingestion.

⚠ Unpublished lecture: 📄 Core RAG Concepts

[+ Content](#)



Understand chunking, embeddings, retrieval, and grounded response generation.

⚠ Unpublished lecture: 📄 Prepare Document Chunks and Metadata [Demo]



[+ Content](#)

Convert extracted document text into retrieval-ready chunks.

⚠ Unpublished lecture: 📄 Generate Embeddings and Store Searchable Knowledge [Demo]



[+ Content](#)

Build the knowledge layer behind the copilot.

⚠ Unpublished lecture: 📄 Retrieve Relevant Content for User Questions [...] ^

+ Content

Select the right document context at runtime.

⚠ Unpublished lecture: 📄 Ground Responses with Retrieved Knowledge [...] ^

+ Content

Improve factuality and business relevance in responses.

⚠ Unpublished lecture: 📄 Show Citations or Source References in the UI [...] ^

+ Content

Increase trust and traceability for end users.

⚠ Unpublished lecture: 📄 Improve Responses with Summarized Context [...] ^

+ Content

Prevent context overload while retaining the most important facts from prior interactions.

⚠ Unpublished lecture: 📄 Section Review + Content ^

Confirm the portal now supports contextual, document-aware, grounded assistance.

+ Curriculum item

Unpublished Section: 📄 Enterprise Actions, Guardrails, and Workflow Orchestration

⚠ Unpublished lecture: 📄 Section Overview + Content ^

Understand how the embedded copilot can support commercial workflows.

⚠ Unpublished lecture: 📄 Reuse Console Plugins in the Web Application ^

+ Content

Carry forward plugin concepts from the console app into the portal.

⚠ Unpublished lecture: 📄 Trigger Business Actions from the Copilot [Demo] ^

+ Content

Support practical actions such as preparing summaries, recommendations, or internal notes.

⚠ Unpublished lecture: 📄 Add Confirmation Gates for Sensitive Operation... ^

+ Content

Require user approval before state-changing actions are executed.

⚠ Unpublished lecture: 📄 Add Authorization Checks for Sensitive Actions ... ^

+ Content

Ensure only properly authorized users can approve, escalate, or modify business records through the copilot.

⚠ Unpublished lecture: 📄 Add Basic Guardrails and Input Validation [Demo] ^

+ Content

Constrain risky behavior and improve reliability in AI-assisted workflows.

⚠ Unpublished lecture: 📄 Log Action Requests and Outcomes [Demo] ^

+ Content

Capture what the assistant suggested, what the user approved, and what actually happened.

⚠ Unpublished lecture: 📄 Build a Request Triage Workflow [Demo] v

[+ Content](#)

⚠ Unpublished lecture: 📄 Build an Approval Preparation Workflow [Demo] ^

[+ Content](#)

Prepare an approval brief that combines portal data, AI summarization, and recommended next steps.

⚠ Unpublished lecture: 📄 Human-in-the-Loop Workflow Design ^

[+ Content](#)

Understand where the user must stay in control and where automation is appropriate.

⚠ Unpublished lecture: 📄 Section Review

[+ Content](#)

^

Confirm the learner can build a commercially relevant AI-enabled web application.

[+ Curriculum item](#)

Unpublished Section: 📄 Observability, Evaluation, and Production Readiness

⚠ Unpublished lecture: 📄 Section Overview

[+ Content](#)

^

Understand why working AI features are not enough for production-ready systems.

⚠ Unpublished lecture: 📄 Why Observability Matters for Semantic Kernel ... ^

[+ Content](#)

Understand what logs, traces, and metrics reveal about copilot behavior, plugin execution, and user flows.

⚠ Unpublished lecture: 📄 Add Semantic Kernel Logging and Telemetry [D... ^

[+ Content](#)

Capture useful runtime details for troubleshooting and operational insight.

⚠ Unpublished lecture: 📄 Add OpenTelemetry for Copilot Requests [Demo] ^

[+ Content](#)

Trace requests through the web app, Semantic Kernel layer, and business actions.

⚠ Unpublished lecture: 📄 Observe Function Calls and Workflow Paths [De... ^

[+ Content](#)

See which plugins were selected, what actions were proposed, and where failures occur.

⚠ Unpublished lecture: 📄 Build a Small Evaluation Set for Key Portal Task... ^

[+ Content](#)

Create repeatable checks for document summarization, grounded Q&A, and request triage quality.

⚠ Unpublished lecture: 📄 Compare Outputs and Improve Prompt and Plu... ^

[+ Content](#)

Use evaluation feedback to improve accuracy, consistency, and user trust.

⚠ Unpublished lecture: 📄 Review Reliability, Cost, and UX Tradeoffs ^

[+ Content](#)

Understand the practical balance between model quality, latency, token usage, and user experience.

⚠ Unpublished lecture: 📄 Prepare the Portal for Deployment [Demo] ^

[+ Content](#)

Finalize configuration, secrets, environment concerns, and deployment readiness.

⚠ Unpublished lecture: 📄 Section Review

+ Content



Confirm the learner can build a commercially credible, observable, and controllable Semantic Kernel solution.

+ Curriculum item

Unpublished Section: 📄 Conclusion and Next Steps

⚠ Unpublished lecture: 📄 Course Recap

+ Content



Review the journey from foundational Semantic Kernel concepts to a web-based embedded copilot.

⚠ Unpublished lecture: 📄 What You Built and What You Learned



+ Content

Connect the final portal implementation to the major technical skills developed in the course.

⚠ Unpublished lecture: 📄 Where to Go Next with Semantic Kernel



+ Content

Identify next-step topics such as more advanced orchestration, memory, enterprise integrations, and agentic systems.

⚠ Unpublished lecture: 📄 Conclusion

+ Content



Close the course with a clear summary and practical encouragement for continued application building.

+ Curriculum item